

Translation — prompts

The translator runs one Gemini API call per non-English (page, TOC section) chunk during the index building step (Gemini 3.1 Flash Lite, no tools): turns the source-language passage into English. The translation feeds two channels — the FTS5 `text_en` column for keyword matching, and the BGE-M3 vector for the ChromaDB `text_chunks` collection — so English queries match English-on-English across both retrieval surfaces. Two prompt surfaces reach the model.

System prompt

Frames the translation task and the preservation rules for technical identifiers (model numbers, part identifiers, standard references, product names).

Placeholders: none (static text).

Template:

```
You translate technical documentation to English.
```

```
Rules:
```

- Preserve technical terms, model numbers, part identifiers, and standard references LITERALLY ("Sensor S-304", "DN25", "DIN 7603", "Klimagerät" when it is a product name). Translate everything else to natural English.
- Output ONLY the translation. No preamble, no quotes, no explanation.

Per-chunk user message

One per non-English (page, TOC section) chunk. Surviving blocks on a page are grouped by their TOC section and merged into one chunk per group; the merged text is sent to the translator. Upstream filtering drops only true noise — empty blocks, table-cell-shaped blocks, and numeric-heavy blocks — with an exemption for list items (numbered steps, bullets) and short ISO-style section headings so procedural lists and headings stay intact.

Placeholders:

- `{text}` — the merged source text of every surviving block under the chunk's TOC section on its page, joined by `\n`.
- `{section_heading}` — optional; the chunk's TOC section heading. Omitted when empty.

Template (no section heading):

```
{text}
```

Template (with section heading):

```
[Section: {section_heading}]
```

```
{text}
```

Expected output

Plain English translation, no JSON wrapper. The caller writes it into the FTS5 `text_en` column and uses the same string to build the BGE-M3 vector for the matching `text_chunks` entry. If translation returns an empty string (English-source doc or the API is unavailable), both channels fall back to the source-language text.

Parameters

- Output cap defaults to 1024 tokens per chunk. Raise via `--translate-max-tokens N` if merged section chunks hit the `translation-truncated` warning.
- Translation runs in parallel across chunks via a thread pool sized by `--workers N` (default 8). Increase to push more requests concurrently if API quota allows; lower if 429s appear.
- English source and other no-translate languages short-circuit before the API call (no API round-trip).
- On output-cap truncation, a structured `translation-truncated` warning is logged with chunk identifier, source language, input size, and the effective cap — the truncated text is still stored, but the log surfaces the silent loss so the user can raise the cap on the next build.

- The system prompt has no prompt-cache marker — the prompt is well below Gemini 3.1 Flash Lite's minimum cacheable prefix, so a marker would silently no-op.